Education Statistics Project

Question

How do primary and lower secondary enrollment rates compare across selected countries, and how does education spending relate to enrollment outcomes from 2010 to 2014?

Data Loading

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:

```
df = pd.read_csv(
    "../datasets/EdStats/EdStatsData.csv",
    low_memory=False
)
df.shape
```

Out[2]:

```
(886930, 70)
```

Data Availability

Define the countries and indicators used in the analysis, then check whether the selected 2010–2014 window has enough coverage for comparison.

In [3]:

```
countries = [
    "Nigeria",
    "Ghana",
    "South Africa",
    "United States",
    "Canada",
    "China",
    "Japan",
    "India",
    "Germany",
    "Brazil"
]

analysis_countries = [
    "Brazil",
    "Germany",
    "Ghana",
    "India",
    "Japan",
    "United States"
]

indicators = [
    "Adult literacy rate, population 15+ years, both sexes (%)",
    "Youth literacy rate, population 15-24 years, both sexes (%)",
    "Adjusted net enrolment rate, primary, both sexes (%)",
    "Adjusted net enrolment rate, lower secondary, both sexes (%)"
]
```

In [4]:

```
project_df = df[
    df["Country Name"].isin(countries)
    &
    df["Indicator Name"].isin(indicators)
]

project_df.shape
```

Out[4]:

```
(40, 70)
```

In [5]:

```
years = [str(y) for y in range(2010, 2015)]

project_df.groupby("Indicator Name")[years].count()
```

Out[5]:

Indicator Name	2010	2011	2012	2013	2014
Adjusted net enrolment rate, lower secondary, both sexes (%)	2	3	3	5	2
Adjusted net enrolment rate, primary, both sexes (%)	5	6	7	7	5
Adult literacy rate, population 15+ years, both sexes (%)	4	3	2	1	2
Youth literacy rate, population 15-24 years, both sexes (%)	4	3	2	1	2

In [6]:

```
spending_df = df[
    df["Indicator Name"].str.contains(
        "expenditure",
        case=False,
        na=False
    )
]

coverage = (
    spending_df
    .groupby("Indicator Name")[["2010","2011","2012","2013","2014"]]
    .count()
    .sum(axis=1)
    .sort_values(ascending=False)
)

coverage.head(20)
```

Out[6]:

```
Indicator Name
Government expenditure on education as % of GDP (%)                                     569
Expenditure on education as % of total government expenditure (%)                       556
Expenditure on tertiary as % of government expenditure on education (%)                 520
Expenditure on secondary as % of government expenditure on education (%)                494
Expenditure on primary as % of government expenditure on education (%)                  490
Government expenditure per primary student as % of GDP per capita (%)                   489
Current expenditure as % of total expenditure in primary public institutions (%)        452
Current expenditure as % of total expenditure in secondary public institutions (%)      448
Government expenditure per tertiary student as % of GDP per capita (%)                  434
Government expenditure per secondary student as % of GDP per capita (%)                 433
Current expenditure as % of total expenditure in public institutions (%)                433
All staff compensation as % of total expenditure in primary public institutions (%)     431
Current expenditure as % of total expenditure in tertiary public institutions (%)       428
All staff compensation as % of total expenditure in secondary public institutions (%)   426
Government expenditure on tertiary education as % of GDP (%)                            417
Government expenditure on secondary education as % of GDP (%)                           412
Expenditure on tertiary as % of total government expenditure (%)                        410
Government expenditure on primary education as % of GDP (%)                             401
Government expenditure in educational institutions as % of GDP (%)                      400
Expenditure on secondary as % of total government expenditure (%)                       399
dtype: int64
```

Enrollment Analysis

Focus on primary and lower secondary enrollment for the countries with usable coverage in the 2010–2014 window.

In [7]:

```
enrollment_df = project_df[
    project_df["Indicator Name"].isin([
        "Adjusted net enrolment rate, primary, both sexes (%)",
        "Adjusted net enrolment rate, lower secondary, both sexes (%)"
    ])
]

enrollment_df[
    ["Country Name", "Indicator Name", "2010", "2011", "2012", "2013", "2014"]
]
```

Out[7]:

	Country Name	Indicator Name	2010	2011	2012	2013	2014
186915	Brazil	Adjusted net enrolment rate, lower secondary, ...	NaN	74.373451	73.733627	82.618088	NaN
186919	Brazil	Adjusted net enrolment rate, primary, both sex...	NaN	95.697182	95.513748	93.779037	94.715057
219900	Canada	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	NaN	NaN
219904	Canada	Adjusted net enrolment rate, primary, both sex...	NaN	NaN	99.477478	99.456741	NaN
241890	China	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	NaN	NaN
241894	China	Adjusted net enrolment rate, primary, both sex...	NaN	NaN	NaN	NaN	NaN
355505	Germany	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	NaN	NaN
355509	Germany	Adjusted net enrolment rate, primary, both sex...	99.612267	99.642090	99.595734	99.847969	99.845421
359170	Ghana	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	50.278381	51.623199
359174	Ghana	Adjusted net enrolment rate, primary, both sex...	NaN	83.850082	82.188591	86.831657	89.312286
414145	India	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	74.614967	NaN
414149	India	Adjusted net enrolment rate, primary, both sex...	96.192810	95.762993	96.984512	97.738869	NaN
447130	Japan	Adjusted net enrolment rate, lower secondary, ...	99.922028	99.857643	99.894012	99.951050	NaN
447134	Japan	Adjusted net enrolment rate, primary, both sex...	99.943512	99.949989	99.948631	99.949997	99.950012
619385	Nigeria	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	NaN	NaN
619389	Nigeria	Adjusted net enrolment rate, primary, both sex...	65.673653	NaN	NaN	NaN	NaN
736665	South Africa	Adjusted net enrolment rate, lower secondary, ...	NaN	NaN	NaN	NaN	NaN
736669	South Africa	Adjusted net enrolment rate, primary, both sex...	NaN	NaN	NaN	NaN	NaN
846615	United States	Adjusted net enrolment rate, lower secondary, ...	86.635513	87.674263	88.243698	88.304207	88.212769
846619	United States	Adjusted net enrolment rate, primary, both sex...	94.795822	93.944641	94.109367	93.825844	94.034531

In [8]:

```
years = ["2010", "2011", "2012", "2013", "2014"]

analysis_df = enrollment_df[
    enrollment_df["Country Name"].isin(analysis_countries)
]

long_df = analysis_df.melt(
    id_vars=["Country Name", "Indicator Name"],
    value_vars=years,
    var_name="Year",
    value_name="Enrollment Rate"
)

long_df = long_df.dropna()

long_df.head()
```

Out[8]:

	Country Name	Indicator Name	Year	Enrollment Rate
3	Germany	Adjusted net enrolment rate, primary, both sex...	2010	99.612267
7	India	Adjusted net enrolment rate, primary, both sex...	2010	96.192810
8	Japan	Adjusted net enrolment rate, lower secondary, ...	2010	99.922028
9	Japan	Adjusted net enrolment rate, primary, both sex...	2010	99.943512
10	United States	Adjusted net enrolment rate, lower secondary, ...	2010	86.635513

In [9]:

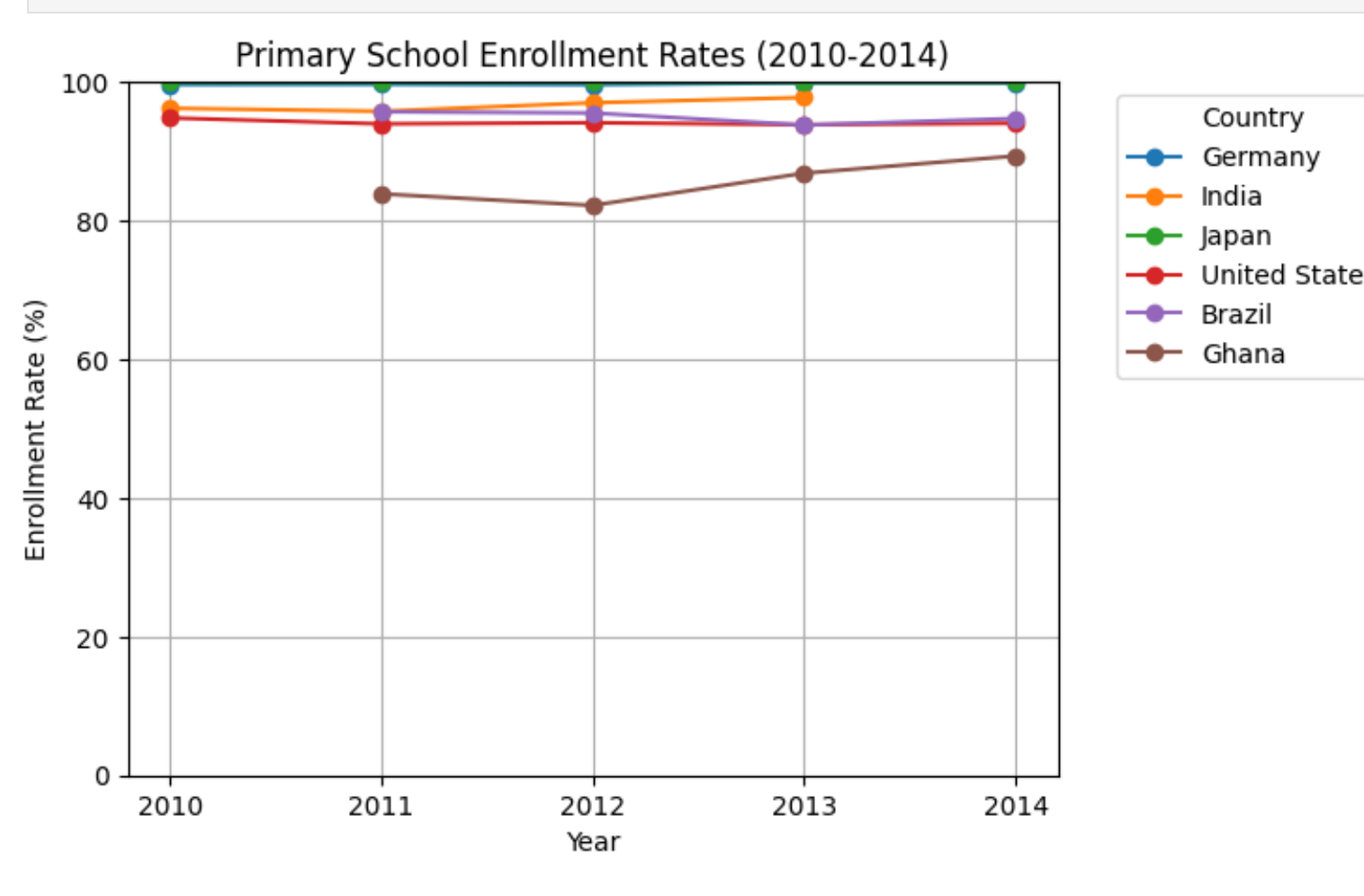
```
primary_df = long_df[
    long_df["Indicator Name"] ==
    "Adjusted net enrolment rate, primary, both sexes (%)"
]

primary_df.head()
```

Out[9]:

	Country Name	Indicator Name	Year	Enrollment Rate
3	Germany	Adjusted net enrolment rate, primary, both sex...	2010	99.612267
7	India	Adjusted net enrolment rate, primary, both sex...	2010	96.192810
9	Japan	Adjusted net enrolment rate, primary, both sex...	2010	99.943512
11	United States	Adjusted net enrolment rate, primary, both sex...	2010	94.795822
13	Brazil	Adjusted net enrolment rate, primary, both sex...	2011	95.697182

In [10]:

```
import matplotlib.pyplot as plt

for country in primary_df["Country Name"].unique():
    country_data = primary_df[
        primary_df["Country Name"] == country
    ]

    plt.plot(
        country_data["Year"],
        country_data["Enrollment Rate"],
        marker="o",
        label=country
    )

plt.title("Primary School Enrollment Rates (2010–2014)")
plt.xlabel("Year")
plt.ylabel("Enrollment Rate (%)")
plt.ylim(0, 100)
plt.legend(
    title="Country",
    bbox_to_anchor=(1.05, 1),
    loc="upper left"
)
plt.grid(True)

plt.grid(True)

plt.savefig(

    "../exports/charts/primary_enrollment.png",

    dpi=300,

    bbox_inches="tight"

)

plt.show()
```



Findings

Primary school enrollment rates remained near universal in Japan and Germany throughout 2010–2014.

India showed steady improvement from approximately 96% to 98%.

Ghana recorded the largest increase, rising from roughly 84% to 89% by 2014.

Brazil and the United States maintained relatively stable enrollment rates around 94–96%.

In [11]:

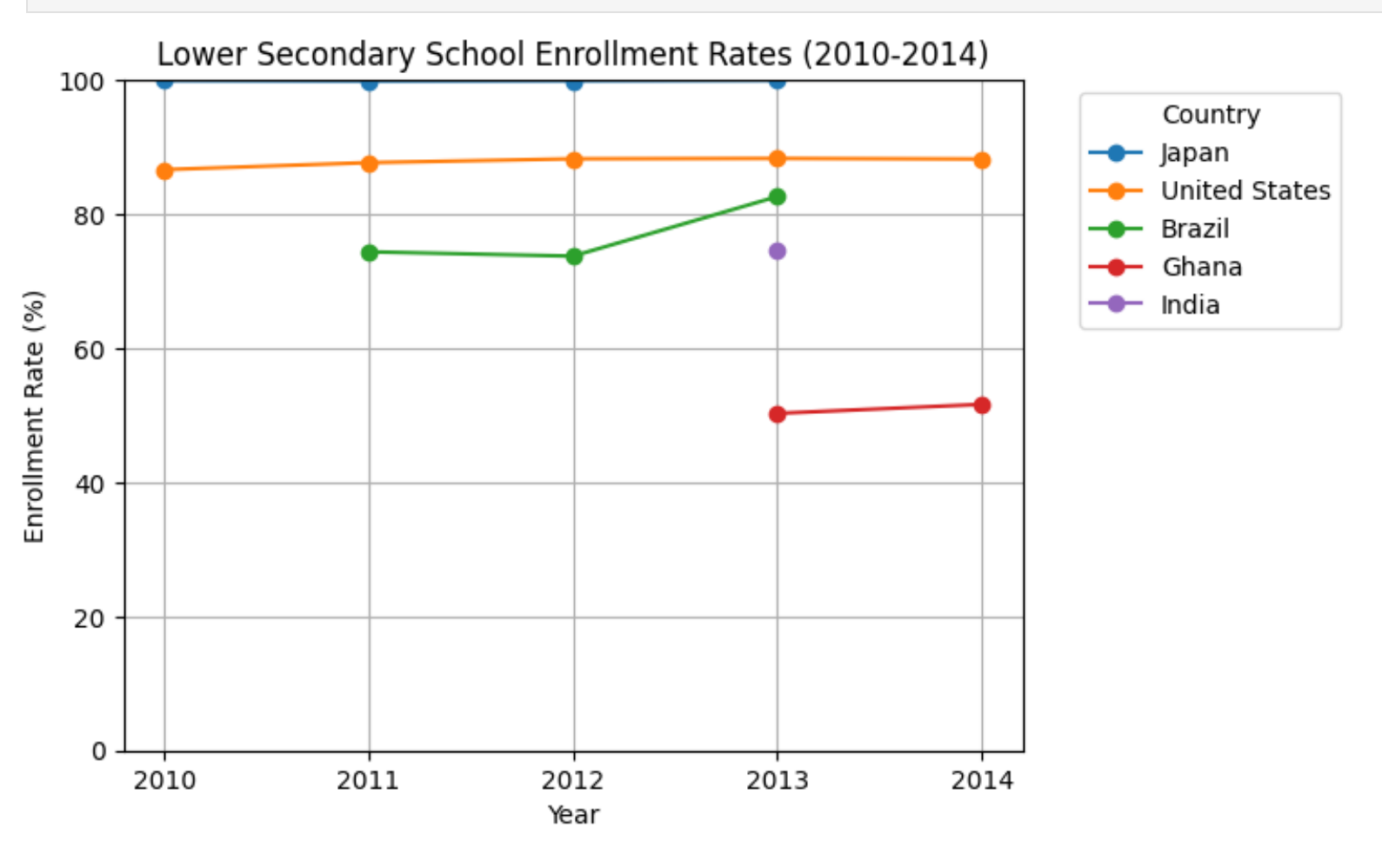
```
secondary_df = long_df[
    long_df["Indicator Name"] ==
    "Adjusted net enrolment rate, lower secondary, both sexes (%)"
]

secondary_df.head()
```

Out[11]:

	Country Name	Indicator Name	Year	Enrollment Rate
8	Japan	Adjusted net enrolment rate, lower secondary, ...	2010	99.922028
10	United States	Adjusted net enrolment rate, lower secondary, ...	2010	86.635513
12	Brazil	Adjusted net enrolment rate, lower secondary, ...	2011	74.373451
20	Japan	Adjusted net enrolment rate, lower secondary, ...	2011	99.857643
22	United States	Adjusted net enrolment rate, lower secondary, ...	2011	87.674263

In [12]:

```
for country in secondary_df["Country Name"].unique():
    country_data = secondary_df[
        secondary_df["Country Name"] == country
    ]

    plt.plot(
        country_data["Year"],
        country_data["Enrollment Rate"],
        marker="o",
        label=country
    )

plt.title("Lower Secondary School Enrollment Rates (2010–2014)")
plt.xlabel("Year")
plt.ylabel("Enrollment Rate (%)")

plt.ylim(0, 100)

plt.legend(
    title="Country",
    bbox_to_anchor=(1.05, 1),
    loc="upper left"
)

plt.grid(True)

plt.savefig(

    "../exports/charts/secondary_enrollment.png",

    dpi=300,

    bbox_inches="tight"

)

plt.show()
```



Secondary Enrollment Findings

While primary school enrollment is high across most countries, larger differences emerge at the secondary level.

Japan maintains nearly universal enrollment throughout the period.

The United States remains relatively stable around 88%.

Brazil improves from approximately 74% to 83%.

Ghana records the lowest secondary enrollment rates, remaining near 50%.

The results suggest that maintaining student participation beyond primary school remains a challenge for some developing economies.

Spending Analysis

Use government expenditure on education as a percentage of GDP to compare spending patterns across the same period.

In [13]:

```
spending_indicator = (
    "Government expenditure on education as % of GDP (%)"
)

spending_df = df[
    (df["Country Name"].isin(countries))
    &
    (df["Indicator Name"] == spending_indicator)
]

spending_df[
    ["Country Name",
     "2010", "2011", "2012", "2013", "2014"]
]
```

Out[13]:

	Country Name	2010	2011	2012	2013	2014
188175	Brazil	5.64277	5.74579	5.79516	5.99395	NaN
221160	Canada	5.36092	5.28122	NaN	NaN	NaN
243150	China	NaN	NaN	NaN	NaN	NaN
356765	Germany	4.91379	4.81336	4.93020	4.92842	4.95219
360430	Ghana	5.54060	8.14097	7.91914	6.09819	6.17855
415405	India	3.42416	3.83866	3.86588	3.84236	NaN
448390	Japan	3.63580	3.64515	3.69533	3.66506	3.59184
620645	Nigeria	NaN	NaN	NaN	NaN	NaN
737925	South Africa	5.71644	5.95843	6.37936	6.02067	6.02955
847875	United States	5.42707	5.21150	5.20087	4.93013	5.38078

In [14]:

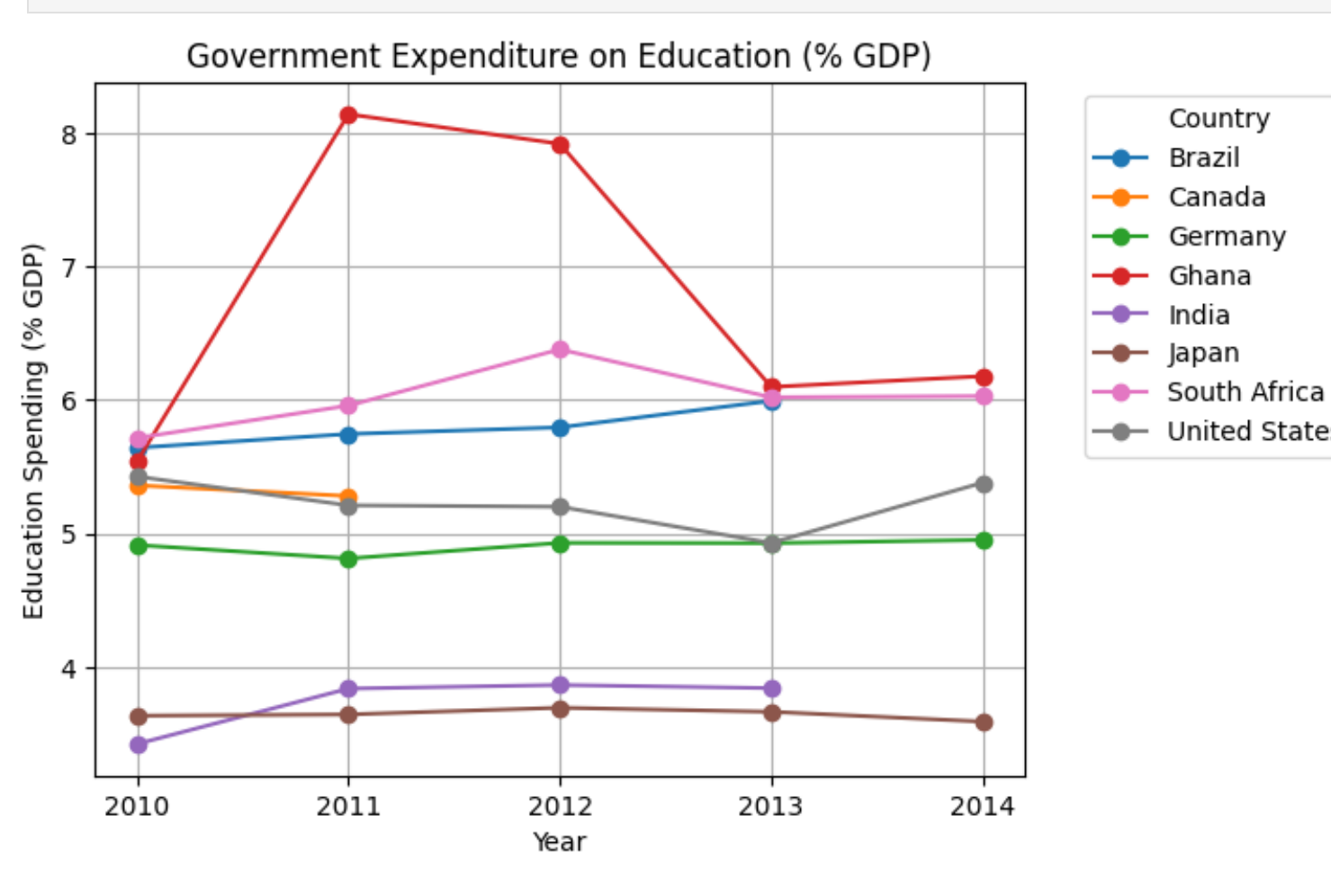
```
spending_long = spending_df.melt(
    id_vars=["Country Name"],
    value_vars=["2010", "2011", "2012", "2013", "2014"],
    var_name="Year",
    value_name="Education Spending (% GDP)"
)

spending_long = spending_long.dropna()

spending_long.head()
```

Out[14]:

	Country Name	Year	Education Spending (% GDP)
0	Brazil	2010	5.64277
1	Canada	2010	5.36092
3	Germany	2010	4.91379
4	Ghana	2010	5.54060
5	India	2010	3.42416

In [15]:

```
for country in spending_long["Country Name"].unique():

    country_data = spending_long[
        spending_long["Country Name"] == country
    ]

    plt.plot(
        country_data["Year"],
        country_data["Education Spending (% GDP)"],
        marker="o",
        label=country
    )

plt.title("Government Expenditure on Education (% GDP)")
plt.xlabel("Year")
plt.ylabel("Education Spending (% GDP)")

plt.legend(
    title="Country",
    bbox_to_anchor=(1.05, 1),
    loc="upper left"
)

plt.grid(True)

plt.savefig(

    "../exports/charts/education_spending_gdp.png",

    dpi=300,

    bbox_inches="tight"

)

plt.show()
```



Spending vs Enrollment

Does spending more on education lead to higher enrollment rates?

To investigate this, we compare:

- Government expenditure on education (% GDP)
- Primary school enrollment rates

for selected countries between 2010 and 2014.

In [16]:

```
primary_avg = (
    primary_df
    .groupby("Country Name")["Enrollment Rate"]
    .mean()
    .round(2)
)

spending_avg = (
    spending_long
    .groupby("Country Name")["Education Spending (% GDP)"]
    .mean()
    .round(2)
)

comparison = pd.concat(
    [primary_avg, spending_avg],
    axis=1
)

comparison
```

Out[16]:

Country Name	Enrollment Rate	Education Spending (% GDP)
Brazil	94.93	5.79
Germany	99.71	4.91
Ghana	85.55	6.78
India	96.67	3.74
Japan	99.95	3.65
United States	94.14	5.23
Canada	NaN	5.32
South Africa	NaN	6.02

In [17]:

```
comparison_clean = comparison.dropna()

comparison_clean
```

Out[17]:

Country Name	Enrollment Rate	Education Spending (% GDP)
Brazil	94.93	5.79
Germany	99.71	4.91
Ghana	85.55	6.78
India	96.67	3.74
Japan	99.95	3.65
United States	94.14	5.23

In [18]:

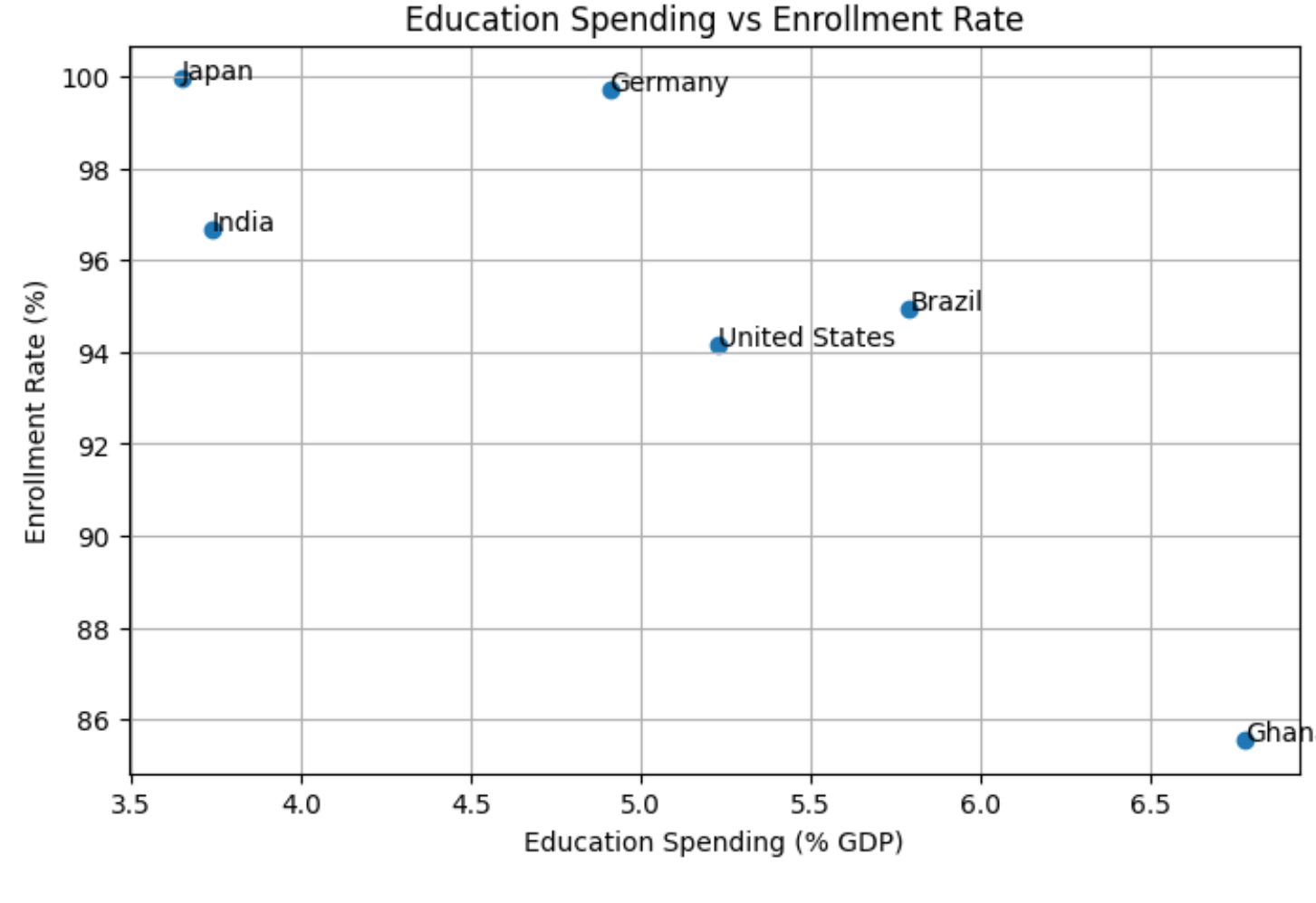
```
comparison_clean.corr(numeric_only=True)
```

Out[18]:

	Enrollment Rate	Education Spending (% GDP)
Enrollment Rate	1.000000	-0.826705
Education Spending (% GDP)	-0.826705	1.000000

In [19]:

```
plt.figure(figsize=(8,5))

plt.scatter(
    comparison_clean["Education Spending (% GDP)"],
    comparison_clean["Enrollment Rate"]
)

for country in comparison_clean.index:
    plt.annotate(
        country,
        (
            comparison_clean.loc[country, "Education Spending (% GDP)"],
            comparison_clean.loc[country, "Enrollment Rate"]
        )
    )

plt.title("Education Spending vs Enrollment Rate")
plt.xlabel("Education Spending (% GDP)")
plt.ylabel("Enrollment Rate (%)")
plt.grid(True)

plt.savefig(

    "../exports/charts/spending_vs_enrollment.png",

    dpi=300,

    bbox_inches="tight"

)

plt.show()
```



Limitations

- Data availability varies by country, year, and indicator.
- Some countries have incomplete lower secondary enrollment or spending data between 2010 and 2014.
- The spending-versus-enrollment comparison uses country averages over a short period.
- The observed correlation does not imply causation.

Conclusion

This project analyzed education indicators from the World Bank EdStats dataset for selected countries between 2010 and 2014.

Key findings include:

- Japan and Germany maintained near-universal primary school enrollment rates.
- Ghana devoted the largest share of GDP to education among the countries analyzed.
- A strong negative correlation (-0.83) was observed between average education spending as a percentage of GDP and average primary school enrollment rates.
- This result does not imply causation and may reflect differences in economic development, educational efficiency, demographics, and policy effectiveness.
- Data availability varied significantly across countries and indicators, limiting the scope of the analysis.

Overall, the analysis suggests that higher education spending alone is not sufficient to explain enrollment outcomes. Additional factors should be considered when evaluating educational performance across countries.